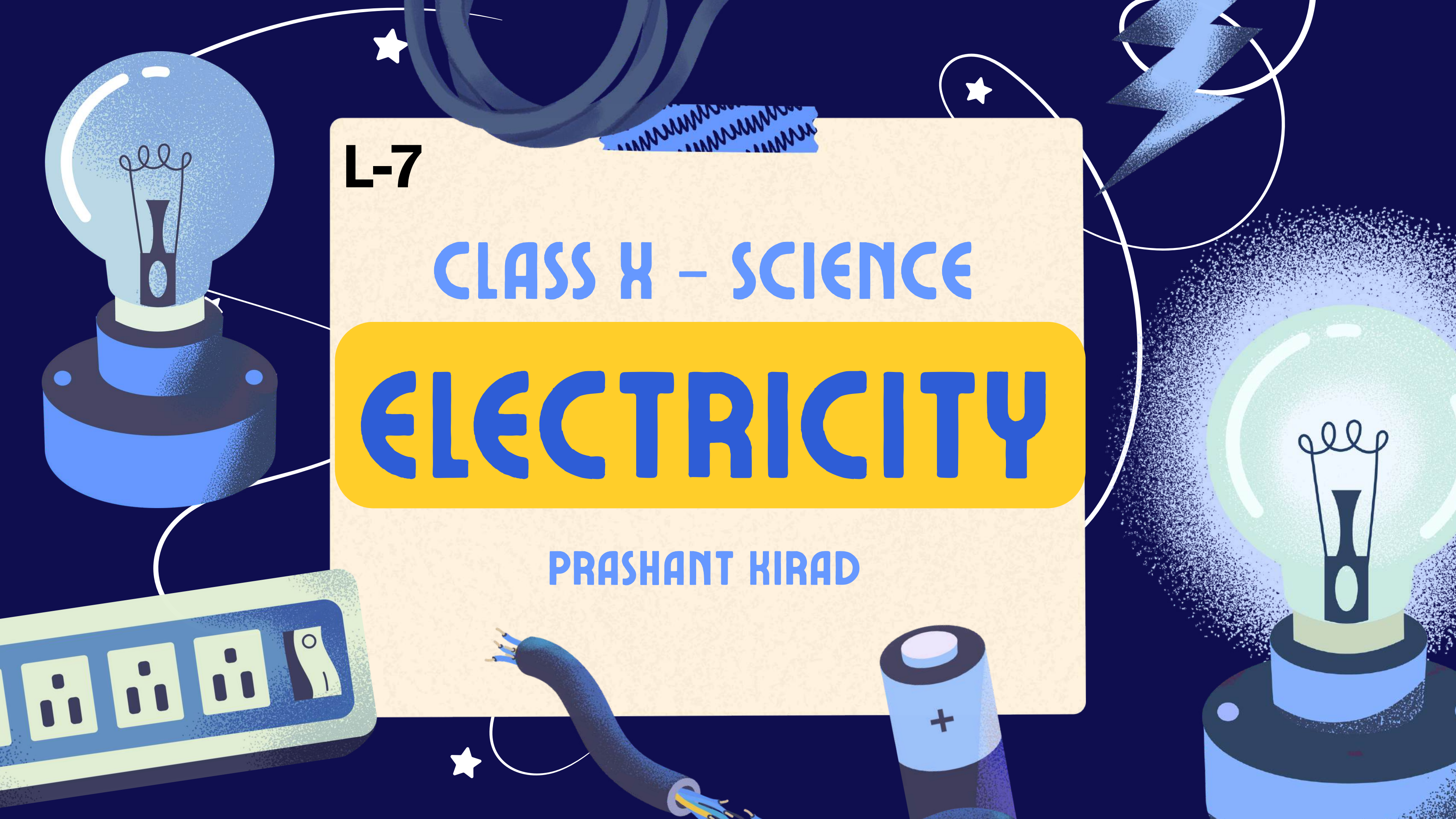


L-7

CLASS X – SCIENCE

ELECTRICITY

PRASHANT KIRAD

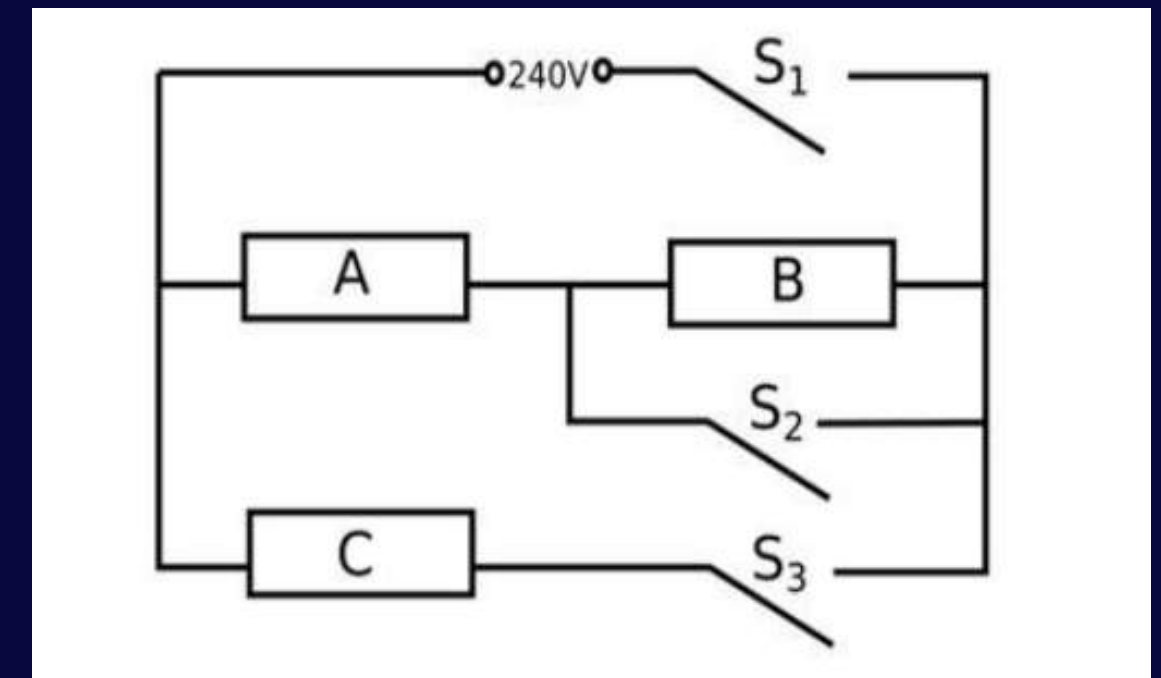


JOSH METER



Q. An electric heater consists of three similar heating elements A, B and C, connected as shown in the figure above. Each heating element is rated as 1.2 kW, 240 V and has constant resistance. S₁, S₂ and S₃ are respective switches. The circuit is connected to a 240 V supply.

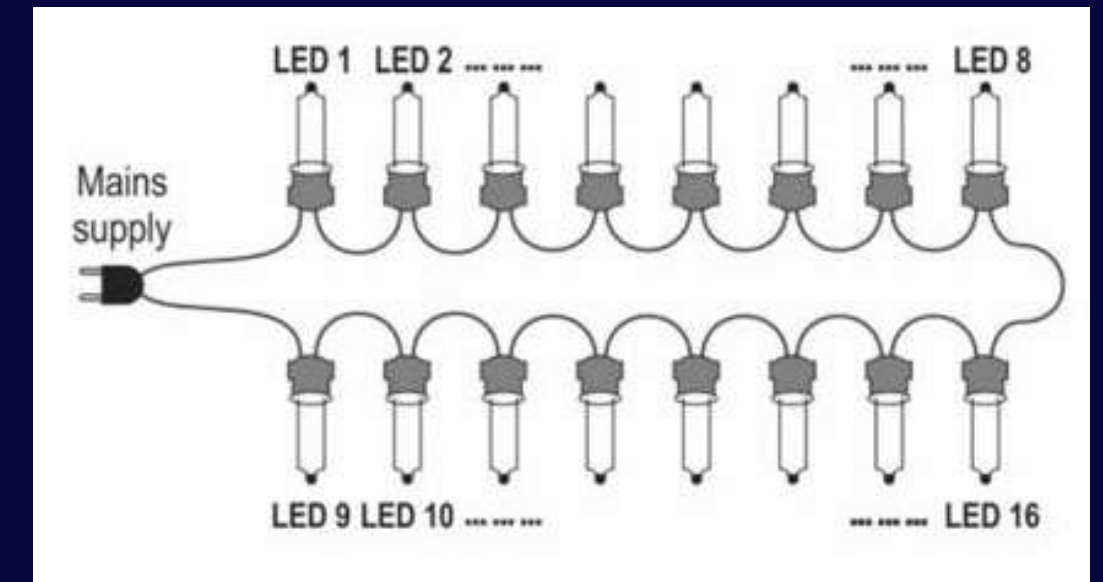
- (i)** Calculate the resistance of one heating element.
- (ii)** Calculate the current in each resistor when only S₁ and S₃ are closed.
- (iii)** Calculate the power dissipated across A when S₁, S₂ and S₃ are closed.



Q. Read the information given below and answer following questions.
Suresh bought a packet of 100 LEDs to make his own lights for decoration in his house. The packet on the LEDs had the following printed on a label:

LED 2835, 0.2 W, 30 Lumens, 3 V

To understand how he should connect the LEDs, he referred to the following circuit diagram on a website.



Q1. Which of the following describes how the LEDs are connected in the circuit diagram?

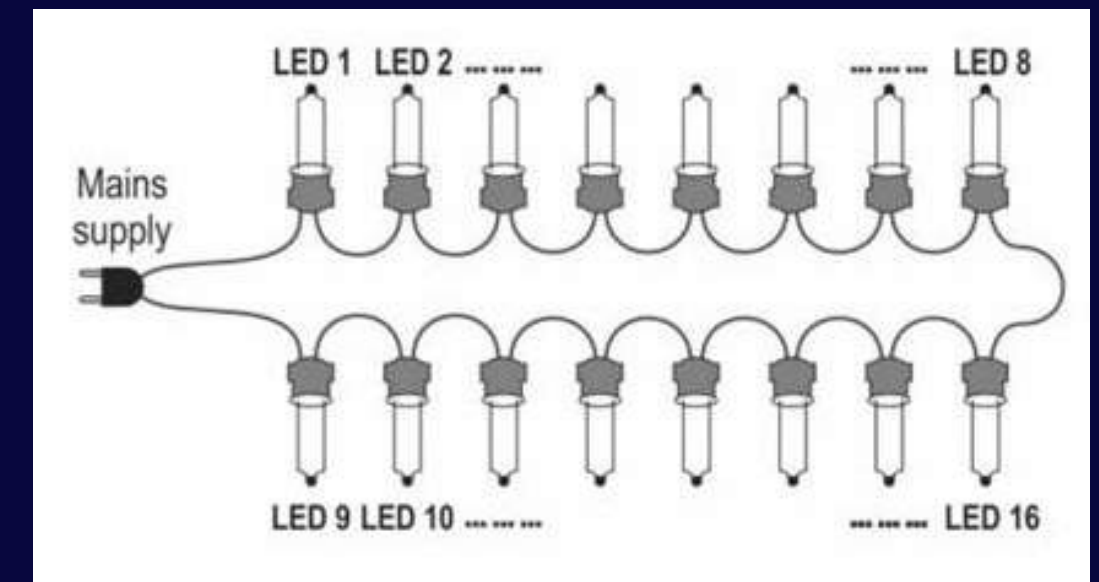
- (A) All in series
- (B) All in parallel
- (C) 8 each in a series combination, and the two combinations in parallel
- (D) 8 each in a parallel combination, and the two combinations in series

Q2. If the LED marked 'LED 2' in the diagram stops working, which other LEDs will also stop working?

- (A) Only LED 3 to LED 8
- (B) Only LED 3 to LED 8 and LED 1
- (C) All the other LEDs in the circuit
- (D) None of the other LEDs in the circuit

Q3. Suresh decided to connect all the LEDs in his lights in a series combination. How many LEDs will he need to connect if he is going to connect the lights to a 240 V mains supply so that the LEDs work at their power rating?

- (A) 16**
- (B) 80**
- (C) 240**
- (D) 1200**



Q4. What will happen if he connects 100 LEDs, all in a parallel combination, to the 240 V mains supply?

- (A) Each LED will work as expected since the available voltage is more than 3 V.**
- (B) Each LED will have a potential difference of 220 V and therefore they will get damaged.**
- (C) Each LED will glow but the ones closer in the circuit to the main supply will glow brighter.**
- (D) Each LED will have a potential difference of 2.4 V across it and therefore will glow dimmer than normal.**

ELECTRIC ENERGY

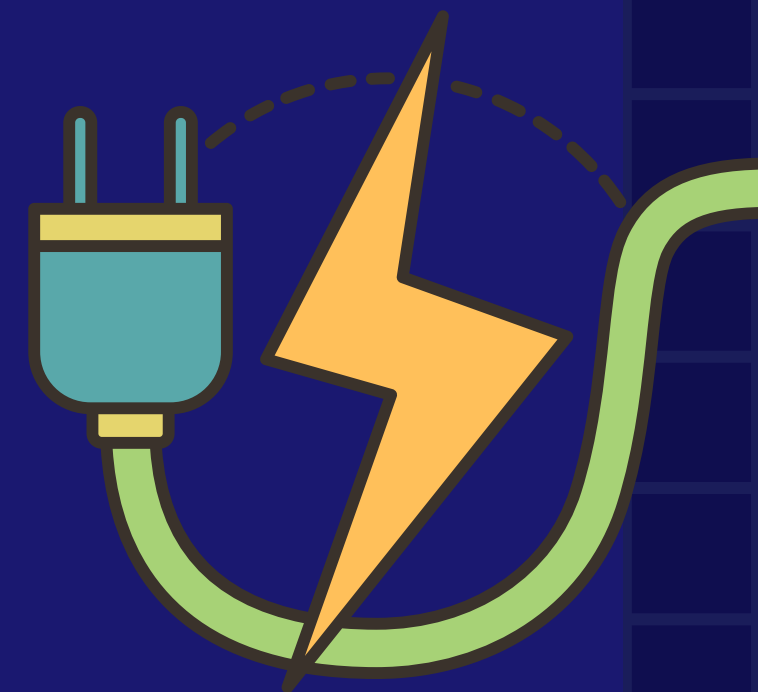
The energy consumed by an electric circuit to flow current through it is referred as electrical energy.

Electrical energy is the product of power and time.

Electrical Energy = Power × Time

$$E = P \times t$$

- The unit of electrical energy is **watt hour (Wh)**.
- **One watt hour** is the energy consumed when 1 watt power is used for 1 hour.



COMMERCIAL UNIT OF ENERGY

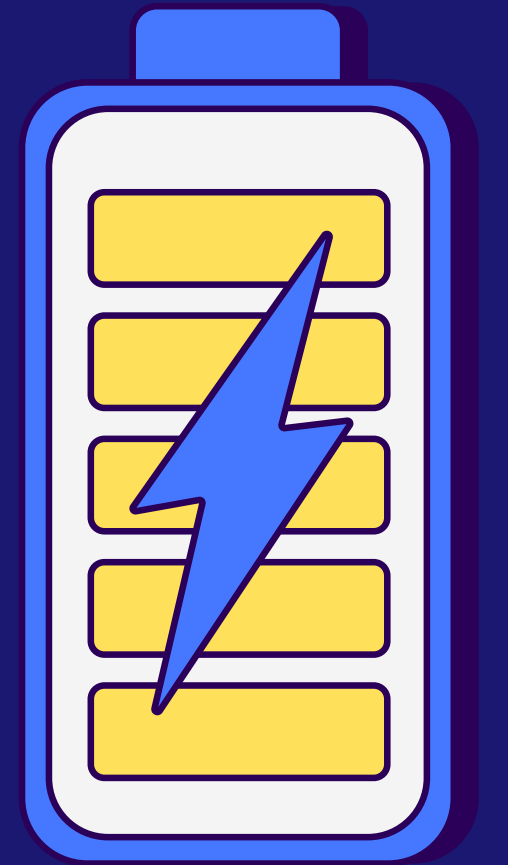
The commercial unit of electrical energy is **kilowatt hour (kWh)**.

It is commonly called 1 unit.

$$1 \text{ kWh} = 1 \text{ unit}$$

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s}$$

$$1 \text{ kWh} = 3.6 \times 10^6$$



[Making Scheme, Delhi, 2020]

Q. An electric bulb is rated at 200 V, 100 W. What is its resistance?

(b) Calculate the electrical energy consumed by 3 such bulbs if they glow continuously for 10 hours daily for the month of November.

(c) Calculate the total cost after the tariff is ₹6.50 per unit.

NCERT

Q. An electric refrigerator rated 400 W operates 8 hour/day. What is the cost of the energy to operate it for 30 days at Rs 3.00 per kW h?

[OEB]

Q. Three 60 W incandescent bulbs are connected in parallel.

(a) Calculate the total power consumed.

(b) Now suppose that one of the bulbs is fused. What will be the total power consumed now?

Q. The commercial unit of electric energy is:

(A) kWh

(B) Watt

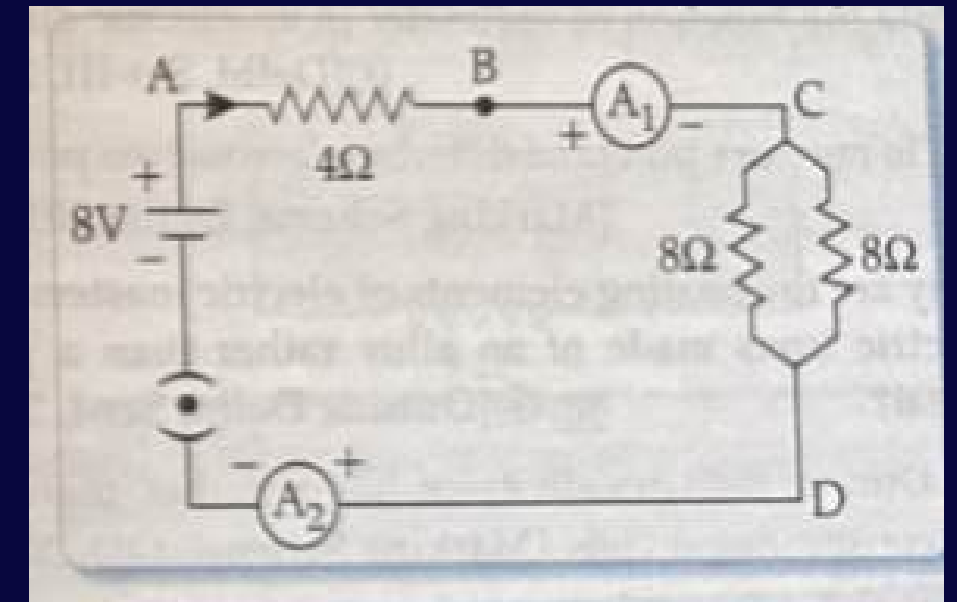
(C) Volt

(D) Volt-ampere

Q. Two identical resistors, each of resistance $15\ \Omega$, are connected in (A) series and (B) parallel, in turn to a battery of 6 V. Calculate the ratio of the power consumed in the combination of resistors in each case.

Q. Find out the following in the electric circuit given:

- (a) Effective resistance of two $8\ \Omega$ resistors in the combination.**
- (b) Current flowing through $4\ \Omega$ resistor.**
- (c) Potential difference across $4\ \Omega$ resistance.**
- (d) Power dissipated in $4\ \Omega$ resistor.**
- (e) Direction of current, if any.**



Q. The arrangement of resistors shown in the given figure is connected to a battery.

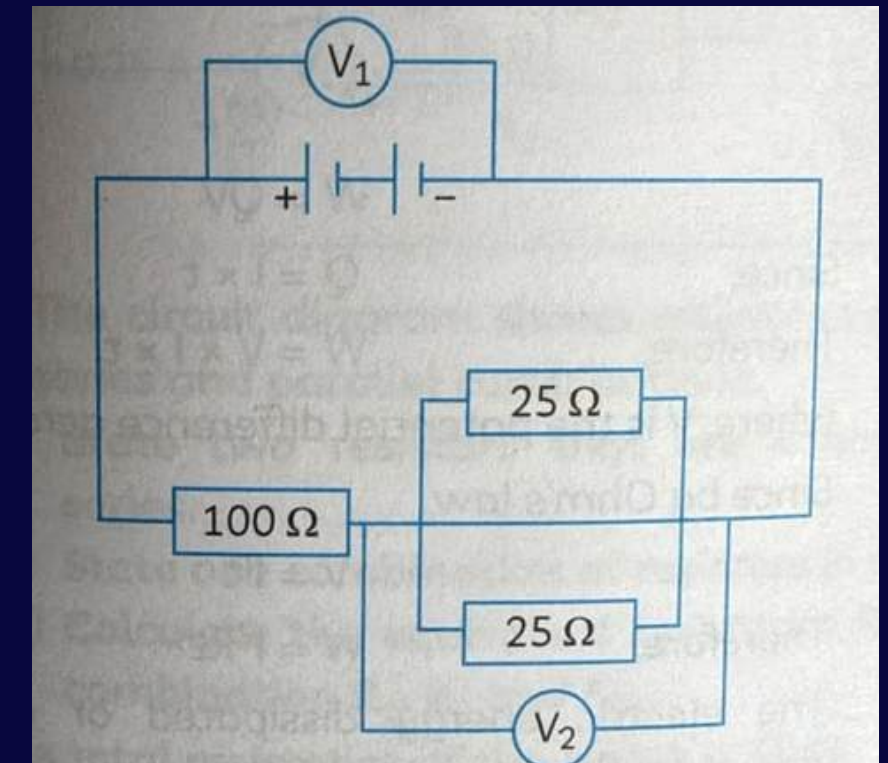
The power dissipation in the $100\ \Omega$ resistor is $81\ \text{W}$.

Calculate:

(A) the current in the circuit

(B) the reading in the voltmeter V_2

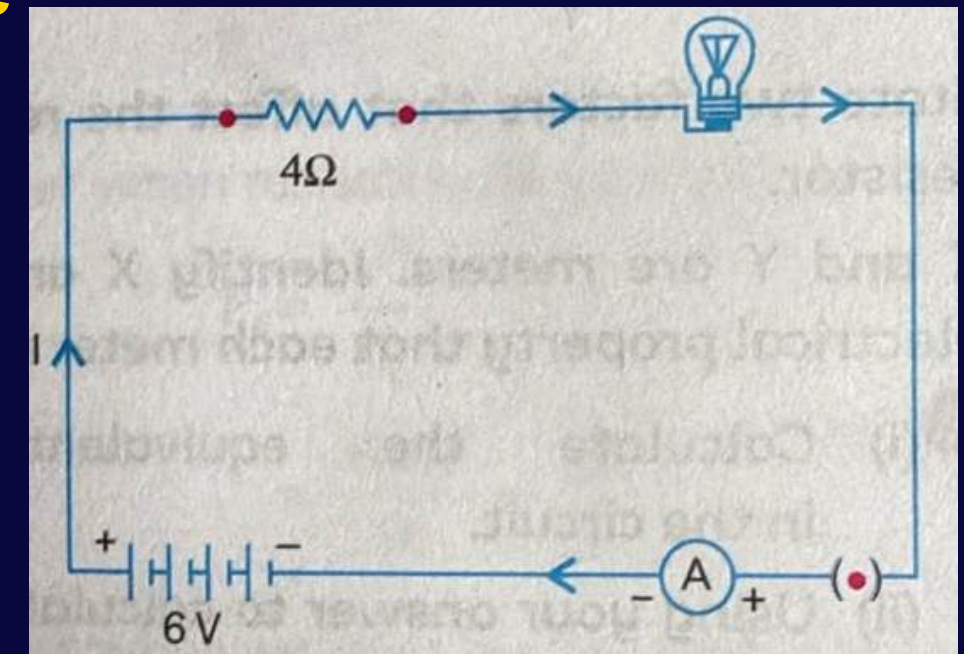
(C) the reading in the voltmeter V_1



Q. An electric lamp of resistance $20\ \Omega$ and a conductor of resistance $4\ \Omega$ are connected to a $6\ \text{V}$ battery as shown in the circuit.

Calculate:

- (A)** the total resistance of the circuit.
- (B)** the current through the circuit.
- (C)** the potential difference across the
 - (i)** electric lamp and
 - (ii)** conductor.
- (D)** power of the lamp.

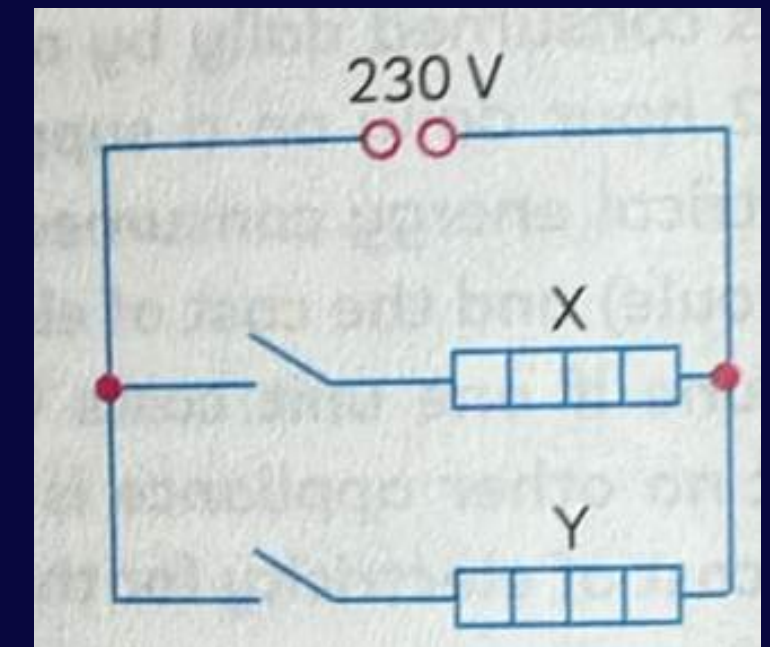


Q. The electric circuit (above figure) in a clothes dryer contains two heaters X and Y in parallel. The above figure shows the circuit connected to a 230 V power supply. When both switches are closed, the current in X is 3.5 A.

Analyse the circuit given above and answer the following questions.

(A) Calculate the power developed in heater X.

(B) If the resistance of X is double that of Y, calculate the current in heater Y.



Chapter Finish!

